

Application of Artificial Intelligence

Unit 3 Consolidation – Individual Activities

Task 1: Timeline Mapping

- **Aristotle's deductive reasoning (circa 350 BC)** - Deductive reasoning is a logical approach based on moving from general ideas to specific conclusions. (For example: All dogs are animals. Bingo is a dog. Therefore, Bingo is an animal.)
This formed a framework for modern understandings of logic and is applicable in the modern study of mathematics and philosophy.
- **Blaise Pascal's Pascaline (1642)** - This was a very early computer capable of mathematical calculations via gear mechanisms. While it could only perform addition and subtraction, it had applications in business and tax, and so was one of the earliest commercial machines.
- **Charles Babbage's Analytical Engine (1837):** This was an early computer featuring both memory and a processor. It was a framework for the architecture of modern computers.
- **George Boole's Law of Thought (1847)** - Boolean algebra was a mathematical framework for expressing logical statements as algebraic equations. This provided a language for encoding propositions as manipulable symbols, allowing the workings and results of deductive reasoning to be more simply and logically recorded.
- **McCulloch-Pitts Artificial Neuron (1943)** - McCulloch and Pitts published a mathematical model of a neuron as a binary threshold logic unit. It began the idea that intelligence could be generated by simple connected units.
- **Von Neumann's Architecture (1945)** - This was a computer architecture that stored processing instructions (programmes) in memory, rather than having them hard-written within the hardware. This meant that computer programmes could be rewritten, making computers reprogrammable and more versatile.
Further, this was an early separation of computer hardware and software. The von Neumann's model included 4 main components: CPU, Memory, Input/Output devices and a shared bus for the transmission of data.
As a downside, the use of a shared bus meant that CPU and memory could not communicate simultaneously, causing the von Neumann bottleneck.

- **Turing Test (1950)** - Alan Turing's paper posed the question 'Can machines think?'. It proposed the Imitation Game (the Turing Test), wherein a machine can be thought to be capable of thought if its answers cannot be distinguished from human responses to the same question.
This paper greatly influenced research into machine learning in the following decades.
- **The Dartmouth Conference (1956)** - This was a summer workshop held at Dartmouth College in the US. The term 'Artificial Intelligence' was first formed, based on the proposal that intelligence could be precisely defined and that a machine could stimulate it.
- **Invention of the Perceptron (1957)** Developed by Frank Rosenblatt, this machine learning algorithm worked with weights to control how much each input affected the output. The weights were then adjusted for classification errors.
The application of weights to stipulate the desired effect of different inputs on outputs is still applicable in modern image recognition software. As such, the Perceptron contributed heavily to modern neural network technology. However, it could not solve non-linearly separable problems like XOR.
- **John McCarthy's LISP** – This was a programming language for symbolic AI. It introduced key concepts like recursion, tree data structures and dynamic typing, which were suited to representing logical expressions.
- **ELIZA (1966)** - This was an early conversational agent that stimulated human conversation by pattern-matching the user input and generating scripted responses that were customised based on the inputs.
It mimicked a psychotherapist and was able to convince users that they were conversing with a real person, generating trust with them. ELIZA highlighted the illusion that could easily be caused by machine learning engines.
- **ALPAC report (1966) and Lighthill report (1973):** These reports, published in the US and UK respectively, evaluated the state of the development of AI systems and produced heavily critical conclusions. They concluded that AI was less reliable, expensive and slow, as well as unable to solve practical problems. These reports discouraged funding towards AI research and led to the first AI winter in the 1970s and 1980s.
- **The First AI Winter (1970s):** This was a dearth of funding and interest in AI in the US and the UK, due to unmet promises. This was mainly caused by the Lighthill and ALPAC reports described above, as well as the Perceptron's inability to solve XOR, which undermined neural network research.

- **Expert Systems (1970s):** There were rule-based programmed encoding human expertise, using thousands of IF-THEN rules. These could be used to diagnose blood infections (MYCIN). They were an example of early commercially successful AI and triggered an AI boom.
These expert systems were a precursor to modern commercial AI systems, such as website chatbots, and AI-powered digital learning platforms, which also use large amounts of data for decision making, helping them to be more flexible and scalable.
- **The Second AI winter (1980s):** This was another decline in interest and funding. Among other reasons, the expert system boom produced mainly AI systems that worked only in limited specialised fields.
Researchers began to use other terms for their field, including machine learning, statistical learning, and pattern recognition. These terms tended to emphasise the statistical and data-driven workings of AI systems, as opposed to just a general notion of intelligence.
- **Deep Blue vs Kasparov (1997):** IBM's Deep Blue chess computer defeated the then chess world champion Garry Kasparov in a series of games played under standard tournament conditions. As opposed to ML, Deep Blue applied a brute-force parallel search algorithm, evaluating about 200 million positions per second to determine optimal moves. This further demonstrated to capabilities of machine learning algorithms.
- **Commercial chatbots (1990s - 2000s):** Rule-based chatbots became common on customer-facing websites to answer customer queries. These started as simple FAQ chatbots that worked by matching keywords. These chatbots improved in the 2010s to include more capable Natural Language Processing (NLP).
- **Pretrained Large Language Models (2018) -** OpenAI's GPT-1 and Google's BERT were released in the same year. Showing that LLMs could be trained for use on a wide range of tasks, as opposed to only specific work.
- **Open AI 's ChatGPT (2022) -** OpenAI's ChatGPT was released in late 2022 and adopted rapidly. It included Reinforcement Learning from Human Feedback (RLHF) which allowed it to adapt to users' preferences, making it more helpful and practical. This also triggered similar releases and adoption of AI solutions by major technology companies.
- **The Frontier Model race including GPT-4, Gemini and Claude (2023 - present) -**

OpenAI's o1/o3 series and DeepSeek's R1 introduced models for extended chain-of-thought reasoning, improving performance on mathematics, coding, and scientific tasks. Different LLMs were released and adopted, as they became more applicable for a wide range of tasks.

- **AI Safety and regulation (2023 - present)** - The widening application of AI solutions has triggered widespread policy debates on AI safety and regulation, including safeguards against harmful usage.

It has also triggered questions about the attribution of Intellectual Property rights for AI-generated content, and who should take responsibility for errors in services provided or assisted by AI-powered systems. This era also marked the spread of services and software to help detect AI-generated content and images, in order to prevent fraud and misinformation.

Reflection:

It is difficult to identify one event with the most significant impact, but any discussion of significant events must include the use of chatbots by websites and the launch of OpenAI's Chat-GPT. Chatbots on commercial websites were an early widespread adoption of an AI solution. However, Chat-GPT has had an even bigger impact in modern perceptions of AI.

For one thing, Chat-GPT was adopted quickly, reaching millions of users in weeks, becoming one of the fastest-adopted electronic products in history. Because of this fast adoption, Chat-GPT has been able to shape modern wide-spread attitudes around AI, almost becoming synonymous with the term itself in some contexts.

Task 2: Symbolic vs. Subsymbolic AI Comparison

Symbolic AI is based on logic and rules understandable by humans such as IF-THEN rules. Examples include medical diagnosis systems which can be programmed to consider a set of symptoms and produce a diagnosis based on hard-coded rules.

As rules are hard-coded, relatively little data is required, and errors can be spotted and corrected quickly. However, these systems are rigid, with the ability to handle only problems they have been designed for.

Subsymbolic systems learn by processing large amounts of data. It adjusts its internal numbers (weights) until it can perform a task with high accuracy. Common examples are image recognition software. The system can learn to recognise an image of a bicycle by viewing several

images of bicycles, noting their features (e.g. handlebars, gears, wheels), and matching these to the images it assesses. Other examples include LLMs and speech assistants.

Subsymbolic systems handle a wider range of tasks than symbolic systems. They can flexibly be taught new tasks but are expensive to create and train. Further, the reasoning behind subsymbolic outputs cannot always be easily explained (the black box problem)

Symbolic intelligence is based on an understanding of processes, while subsymbolic intelligence is based on experience.

These paradigms complement each other through the application of neuro-symbolic AI, which combines the competencies of both systems.

Task 3: AI winter reflection

Reasons for AI winters

- **Overestimation of the capabilities of AI and the speed of development:** The main reasons were that early discourse of AI overestimated its capabilities. As such, AI systems did not develop fast enough to keep up with expectations. Funding for expensive research soon reduced, leading to even slower progress.
- **Impact of the Lighthill and ALPAC reports:** The Lighthill report (1973) and the ALPAC report (1966) were heavily critical of the time and cost investment in AI development projects, which led to reduced enthusiasm and lower funding.
- **Reduced funding:** lower availability of funding from bodies like universities and governments reduced the research that could be done into AI solutions.

Lessons for today's AI researchers and companies:

- **The Importance of managing expectations:** It is vital for researchers to be realistic about what AI systems can be expected to do. It is also important for companies to entrust roles only after the systems have been extensively tested and confirmed to perform the task adequately.
- **The effect of rebranding:** Rebranding AI research as machine learning and statistical research presented a way for researchers to access funding during AI winters. Effective branding and naming will affect a product's reception.

- **Importance of defining AI's potential more clearly:** As Floridi notes, AI solutions can either replace, diversify, or complement current solutions. It is well possible for AI-powered and human-powered solutions to co-exist, and so investment in AI-powered solutions should not be at the expense of the development of other solutions.

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